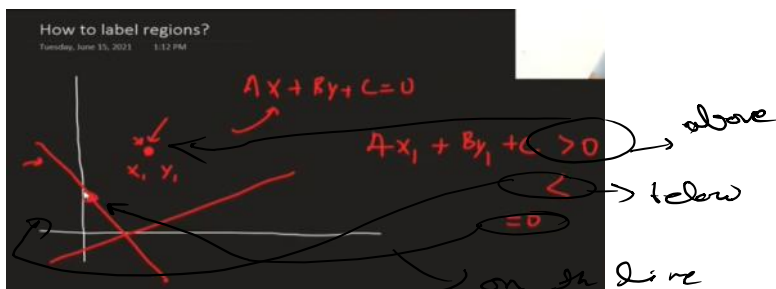
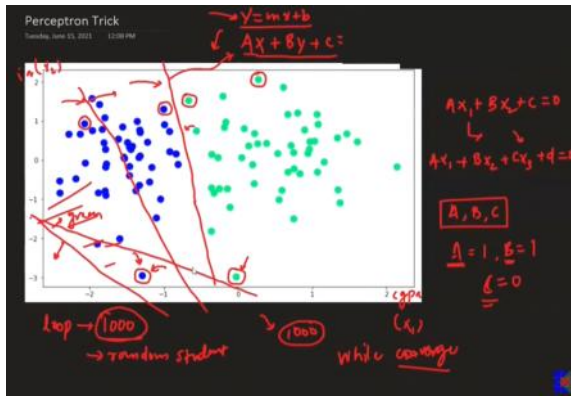
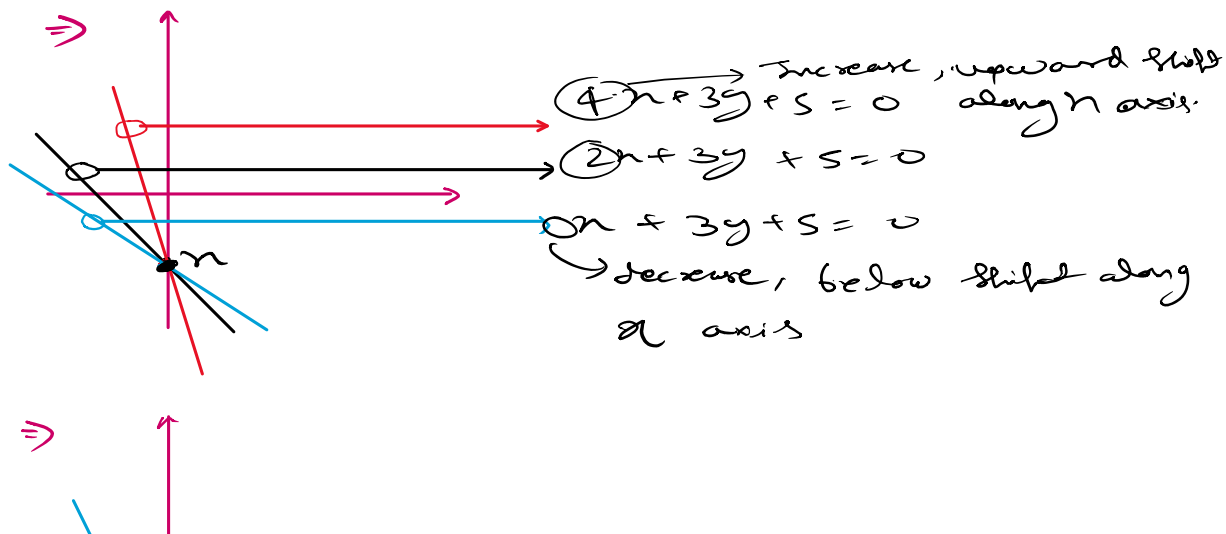
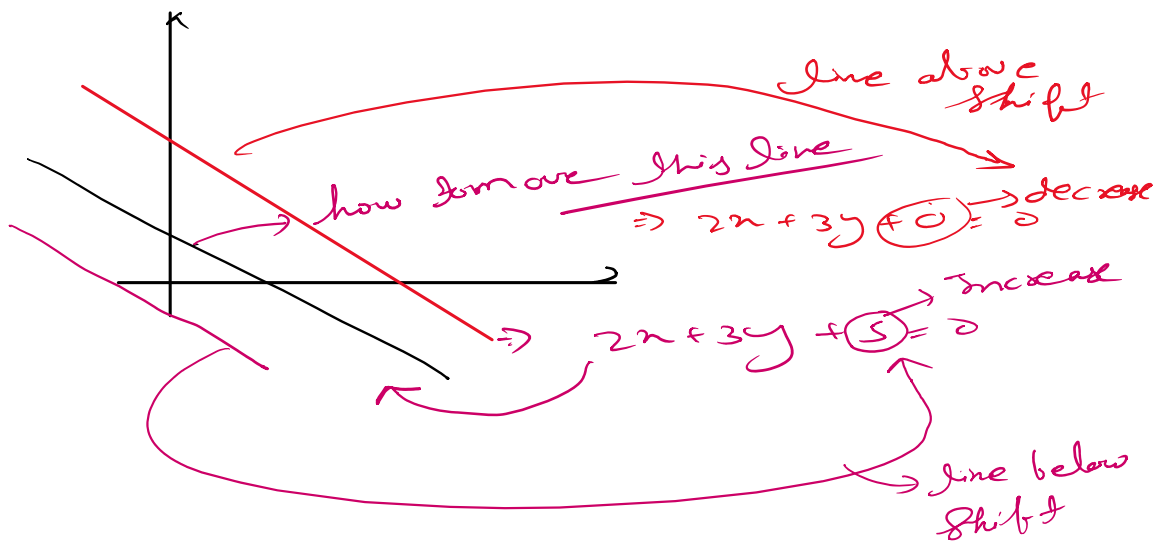


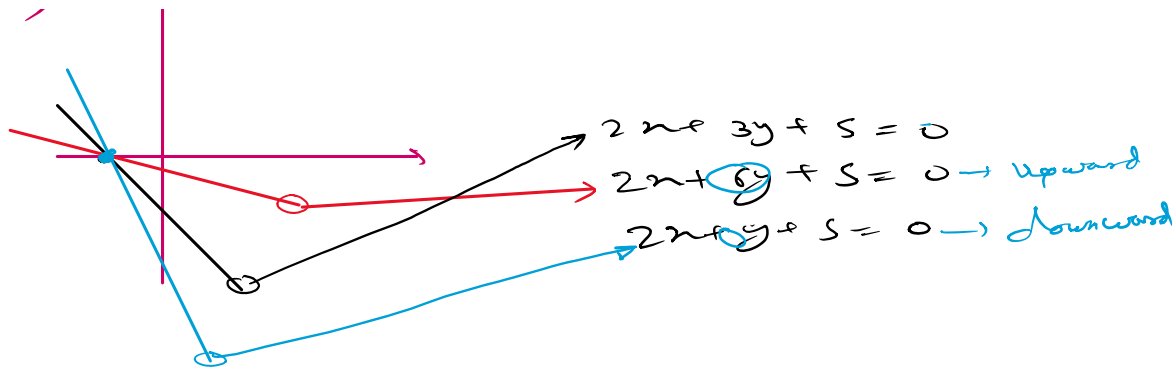
Perceptron Trick

Sunday, September 6, 2026 1:10 AM

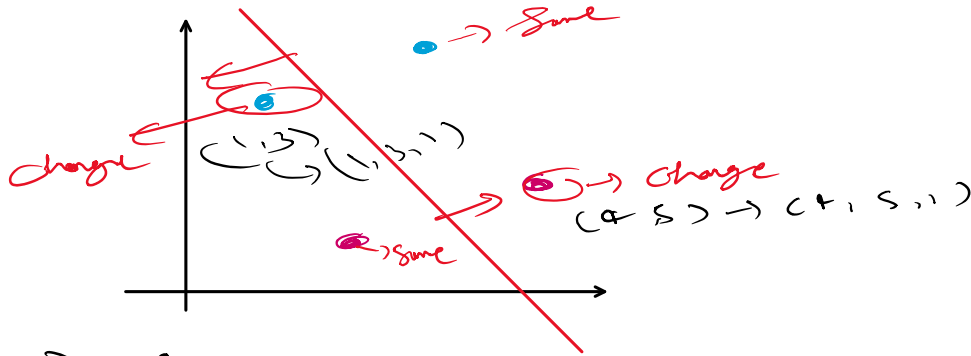


$\Rightarrow$ How to Shift line





⇒ Transformation: —



⇒ Algorithm: —

(x_1)	(x_2)	$\Rightarrow$
7.5	81	1
8.5	105	1
7.0	81	0

$$Ax + By + C = 0$$

$$w_0 + w_1x_1 + w_2x_2 = 0$$

$$C, w_1 = A, w_2 = B$$

$$w_0x_0 + w_1x_1 + w_2x_2 = 0$$

$$\sum_{i=0}^n w_i x_i = 0$$

$w_0x_0 + w_1x_1 + w_2x_2 \geq 0 \rightarrow \text{placement high}$
 $\leq 0 \rightarrow \text{placement nhi chaga}$

⇒ algo: —

epoch $\rightarrow 1000, \eta = 0.01$

for i in range (epoch):

randomly select a student

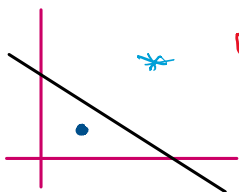
if $x_i \in N$ and $\sum_{i=0}^n w_i x_i \geq 0$ placement high

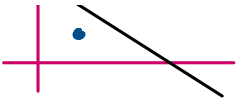
placement nhi chaga

$w_{\text{new}} = w_{\text{old}} - \eta x_i \rightarrow \text{update}$

if $x_i \in P$ and $\sum_{i=0}^n w_i x_i < 0$

placement o chaga





$-1 \leq w \leq 1$

$z=0$

$w_{new} = w_{old} + \eta x_i$ → update
placement nhi hua

⇒ Simplified Algo:- *Simplified :-*

if $x_i \in N$ and $\sum w_i x_i \geq 0$
 $w_\eta = w_0 - \eta x_i$
 if $x_i \in P$ and $\sum w_i x_i < 0$
 $w_\eta = w_0 + \eta x_i$

for i in 1000;
 random student
 $w_\eta = w_0 + \eta (y_i - \hat{y}_i) x_i$

model predict

y_i <i>actual</i>	$\hat{y}_i$	$y_i - \hat{y}_i$
0	1	0
1	0	1
0	1	-1

for i in range(epochs):
 select a random student (i)
 $w_\eta = w_0 + \eta (x_i - \hat{y}_i) x_i$

final algo